

Justification

Building-Age Classification: Madrid to Amsterdam transfer | Team 1: [Kathoni, Brian, Lydia, Peter, Ignatius]

Model design

Inputs are 60 preprocessed spectral and temporal features plus 19 engineered band ratios, band differences and urban-index contrasts; three of the five most important features are engineered. Missing values are median-imputed with Madrid statistics only. Stage 1 is XGBoost (300 trees, depth 8, learning rate 0.04), chosen because it handles correlated tabular features without scaling and can resume boosting from a trained model. Madrid 5-fold CV macro-F1 is 0.639 ± 0.005 , against 0.628 ± 0.004 for the Random Forest baseline.

Transfer strategy

Stage 2 keeps the Madrid booster and adds 50 trees fitted only to n labelled Amsterdam pixels per class. The local labels therefore correct the Madrid model instead of being outweighed by 76,000 Madrid pixels. At 25 pixels per class this scored 0.65, against 0.55 for retraining XGBoost and 0.56 for HistGradientBoosting on pooled data. Adding more than about 50 trees did not help. Everything is seeded and n is a parameter.

Interpretation

Zero-shot F1 is 0.506, with recall of only 0.18 for the 1984–2004 class. Amsterdam F1 for $n = 5, 10, 25, 50, 100, 200$ is 0.585, 0.647, 0.652, 0.673, 0.685: most of the gain arrives by 25 labels per class, then slows. Class 2 (1945–1984) stays hardest, plausibly because its boundary differs between cities. Caveats: Madrid CV and Amsterdam draws use random pixel splits, so neighbouring pixels can leak and scores are optimistic; each Amsterdam std comes from three draws.